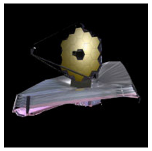

9078 - Probing the origin and interiors of Jupiter Trojans through the study of collisional fragments

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Ian Wong (PI)	**Space Telescope Science Institute**
Dr. Michael E Brown (CoI)	California Institute of Technology
Dr. Joshua P Emery (CoI)	Northern Arizona University
Dr. Keith S. Noll (CoI)	NASA Goddard Space Flight Center
Matthew Belyakov (CoI)	California Institute of Technology

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NIRSpec observations				
	1	ANIUS	NIRSpec IFU Spectroscopy	(1) ANIUS
	2	EURYMACHOS	NIRSpec IFU Spectroscopy	(2) EURYMACHOS
	3	ZAREX	NIRSpec IFU Spectroscopy	(3) ZAREX
	4	ENNOMOS [phase 0]	NIRSpec IFU Spectroscopy	(4) ENNOMOS
	5	ENNOMOS [phase 0.5]	NIRSpec IFU Spectroscopy	(4) ENNOMOS
	6	HIPPASOS	NIRSpec IFU Spectroscopy	(5) HIPPASOS

ABSTRACT

Jupiter Trojans have been a persistent enigma for planetary science. Current models of solar system evolution predict that they must have formed beyond the primordial orbits of the ice giants alongside the present-day Kuiper belt objects. Therefore, detailed study of the surface properties of Trojans should, in principle, reveal the telltale signs of an outer solar system origin. Collisions offer a unique perspective on the question of Trojan

composition. The resurfaced material on collisional fragments provide a window into the bulk composition of these objects and may retain some chemical signatures that are otherwise obscured or degraded on the uncollided objects. The collisional fragment Eurybates was observed by NIRSpec as part of a Cycle 1 JWST program. The spectrum reveals an exceptionally deep 3 micron absorption band, as well as a never-before-seen 4.25 micron feature that suggests the ubiquitous presence of carbon dioxide in the interior of Trojans. Motivated by this result, we propose to collect spectroscopic observations of additional Trojan collisional fragments. We plan to observe 3 smaller members of the Eurybates family to assess whether the 4.25 micron feature on Eurybates is representative of the entire parent body. We will also target 2 members of the Ennomos collisional family to probe whether there are systematic differences in bulk composition among the Trojan population. Rotationally resolved spectra of Ennomos will explore the surface variability apparent in previous ground-based observations.

OBSERVING DESCRIPTION

We will observe 5 Trojan collisional family members with the NIRSpec IFU. The targets are the Eurybates family members 8060 Anius, 9818 Eurymachos, 18060 Zarex, and the Ennomos family members 4709 Ennomos and 17492 Hippasos. Ennomos is known to show rotational variability, and we will observe it twice, targeting opposite hemispheres near 0 and 0.5 rotational phase, in order to probe for compositional graients across the surface that may indicate the presence of both original surface regolith and collisionally resurfaced interior material.

To probe the attested 2.8-3.5 and 4.25 micron features seen in the previous JWST spectrum of Eurybates, we will use the G235M and G395M gratings to provide continuous coverage from 1.7 to 5.3 microns. We require SNR = 40 at 4.25 microns to adequately detect and measure the shape of the carbon dioxide feature; our target of SNR = 46 at 3.0 microns for the G235M observations will ensure consistent sensitivity across the overlapping region between the two grating settings and provide exquisite constraints on the depth and shape of the broad 3 micron band.

Our observing strategy follows standard procedures for faint moving target observations. For each target and grating setting, we will collect a set of dithered single-integration exposures, utilizing the NRSIRS2RAPID readout and a 4-point dither pattern to minimize readout noise and detector effects.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	ANIUS	TYPE=ASTEROID,A=5.194259188089585,E=0.0913211637393089,I=7.083235175513856,O=6.331354494656668,W=33.39965104833954,M=282.1592348625136,EQUINOX=J2000,EPOCH=15-SEP-2018:00:00:00,EpochTimeScale=TDB		
Comments: Eurybates family *Extended=NO*				
(2)	EURYMACHOS	TYPE=ASTEROID,A=5.207169653796187,E=0.00223380701168595,I=7.490262971257319,O=18.58966733565464,W=114.5006727580149,M=151.956009413395,EQUINOX=J2000,EPOCH=14-AUG-2018:00:00:00,EpochTimeScale=TDB		
Comments: Eurybates family *Extended=NO*				
(3)	ZAREX	TYPE=ASTEROID,A=5.12701817208954,E=0.05751585102422171,I=6.637753712112503,O=224.9067122018968,W=152.2518250649314,M=283.5077654448017,EQUINOX=J2000,EPOCH=02-NOV-2018:00:00:00,EpochTimeScale=TDB		
Comments: Eurybates family *Extended=NO*				
(4)	ENNOMOS	TYPE=ASTEROID,A=5.257606989985684,E=0.02112823699299943,I=25.4457278357235,O=253.1198842352379,W=78.41478578597936,M=276.5664221093813,EQUINOX=J2000,EPOCH=12-MAY-2020:00:00:00,EpochTimeScale=TDB		
Comments: Ennomos family *Extended=NO*				
(5)	HIPPASOS	TYPE=ASTEROID,A=5.149685841271473,E=0.06795934625369961,I=29.20134939227213,O=89.00317552084114,W=217.7903135844005,M=277.7981029196394,EQUINOX=J2000,EPOCH=09-MAY-2019:00:00:00,EpochTimeScale=TDB		
Comments: Ennomos family *Extended=NO*				

Observation

Proposal 9078, Observation 1: ANIUS

Wed Nov 19 16:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(ANIUS (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	ANIUS	TYPE=ASTEROID,A=5.194259188089585,E=0.0913211637393089,I=7.083235175513856,O=6.331354494656668,W=33.39965104833954,M=282.1592348625136,EQUINOX=J2000,EPOCH=15-SEP-2018:00:00:00,EpochTimeScale=TDB		

Comments: Eurybates family
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2RAPID	48	1	false	true	NONE	4	4	2859.422	
2	G235M/F170LP	NRSIRS2RAPID	52	1	false	true	NONE	4	4	3092.845	

Special Requirements

DEFAULT WINDOW: ANGULAR RATE ANIUS FROM JWST LESS THAN 0.075

Observation

Proposal 9078, Observation 2: EURYMACHOS

Wed Nov 19 16:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(EURYMACHOS (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(2)	EURYMACHOS	TYPE=ASTEROID,A=5.207169653796187,E=0.00223380701168595,I=7.490262971257319,O=18.58966733565464,W=114.5006727580149,M=151.956009413395,EQUINOX=J2000,EPOCH=14-AUG-2018:00:00:00,EpochTimeScale=TDB		

Comments: Eurybates family
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2RAPID	37	1	false	true	NONE	4	4	2217.511	
2	G235M/F170LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	

Special Requirements

DEFAULT WINDOW: ANGULAR RATE EURYMACHOS FROM JWST LESS THAN 0.075

Observation

Proposal 9078, Observation 3: ZAREX

Wed Nov 19 16:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(ZAREX (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(3)	ZAREX	TYPE=ASTEROID,A=5.12701817208954,E=0.05751585102422171,I=6.637753712112503,O=224.9067122018968,W=152.2518250649314,M=283.5077654448017,EQUINOX=J2000,EPOCH=02-NOV-2018:00:00:00,EpochTimeScale=TDB		

Comments: Eurybates family
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2RAPID	48	1	false	true	NONE	4	4	2859.422	
2	G235M/F170LP	NRSIRS2RAPID	52	1	false	true	NONE	4	4	3092.845	

Special Requirements

DEFAULT WINDOW: ANGULAR RATE ZAREX FROM JWST LESS THAN 0.075

Observation

Proposal 9078, Observation 4: ENNOMOS [phase 0]

Wed Nov 19 16:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(ENNOMOS [phase 0] (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(4)	ENNOMOS	TYPE=ASTEROID,A=5.257606989985684,E=0.02112823699299943,I=25.4457278357235,O=253.1198842352379,W=78.41478578597936,M=276.5664221093813,EQUINOX=J2000,EPOCH=12-MAY-2020:00:00:00,EpochTimeScale=TDB		

Comments: Ennomos family
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2RAPID	6	1	false	true	NONE	4	4	408.489	
2	G235M/F170LP	NRSIRS2RAPID	7	1	false	true	NONE	4	4	466.844	

Special Requirements

Phase -0.1 to 0.1 with period 12.26884 Hours and zero-phase 2459409.81217 HJD

DEFAULT WINDOW: ANGULAR RATE ENNOMOS FROM JWST LESS THAN 0.075

Observation

Proposal 9078, Observation 5: ENNOMOS [phase 0.5]

Wed Nov 19 16:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(ENNOMOS [phase 0.5] (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(4)	ENNOMOS	TYPE=ASTEROID,A=5.257606989985684,E=0.02112823699299943,I=25.4457278357235,O=253.1198842352379,W=78.41478578597936,M=276.5664221093813,EQUINOX=J2000,EPOCH=12-MAY-2020:00:00:00,EpochTimeScale=TDB		

Comments: Ennomos family
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2RAPID	6	1	false	true	NONE	4	4	408.489	
2	G235M/F170LP	NRSIRS2RAPID	7	1	false	true	NONE	4	4	466.844	

Special Requirements

Phase 0.4 to 0.6 with period 12.26884 Hours and zero-phase 2459409.81217 HJD

DEFAULT WINDOW: ANGULAR RATE ENNOMOS FROM JWST LESS THAN 0.075

Observation

Proposal 9078, Observation 6: HIPPASOS

Wed Nov 19 16:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(HIPPASOS (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(5)	HIPPASOS	TYPE=ASTEROID,A=5.149685841271473,E=0.06795934625369961,I=29.20134939227213,O=89.00317552084114,W=217.7903135844005,M=277.7981029196394,EQUINOX=J2000,EPOCH=09-MAY-2019:00:00:00,EpochTimeScale=TDB		

Comments: Ennomos family
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2RAPID	19	1	false	true	NONE	4	4	1167.111	
2	G235M/F170LP	NRSIRS2RAPID	21	1	false	true	NONE	4	4	1283.822	

Special Requirements

DEFAULT WINDOW: ANGULAR RATE HIPPASOS FROM JWST LESS THAN 0.075